

Roll for the Galaxy: Phase Battery Variant

Current Active Direction

The active direction has been backed up to **minimal no-roll Roll**:

- keep secret phase selection;
- in solo and 2p games, each player selects two eligible phases; at 3p+, each player selects one eligible phase;
- a phase is eligible only if the player has at least one ready pip for that phase;
- selecting a phase does not consume pips; pips are workers spent only when a selected phase resolves;
- resolve only selected phases, in normal Roll order;
- restore the VP chip pool and VP-chip scoring;
- restore Ship as Trade-or-Consume;
- use White as the Settle track for non-Military Worlds; Credits are unlimited chips like base Roll;
- remove 6-cost Developments from the normal bag and use them as delayed end-game goal cards;
- do not use the larger all-phases/no-VP-pool/Development-reward branch.

The goal is to remove dice-roll swing while staying close to base Roll for the Galaxy. Some older sections below still describe the larger Phase Battery expansion branch and should be treated as parked until rewritten.

This variant replaces face-roll luck with deterministic phase capacity. Players still build a tableau of Worlds and Developments, pay Galactic Credits, Produce Goods, Ship Goods to Trade for Credits or Consume for VP chips, and score from tableau VP, VP chips, and chosen 6-cost Development goals. The big change is that dice colors become phase tracks instead of rolled workers. Secret phase selection is constrained by available pips, but selecting a phase does not spend those pips.

Core rule:

When a tile grants a die, increase that color's max. Its printed location determines when the new pip becomes ready.

There is no Roll Dice step, no Assign Dice step, no Dice Cup, and no Citizenry.

Current Design Decision

The recommended minimal ruleset is:

- dice colors become deterministic phase batteries;
- each track has current pips and max pips;
- gaining a die increases that color's max; Cup, Citizenry, and World placement determine readiness;
- Red pips are Military value/readiness;
- White pips are Settle workers for non-Military Worlds;
- Military Worlds require enough Red and exhaust 1 current Red after settling;
- normal Worlds use White pips;
- Credits are Trade income and can recharge spent colored pips during Manage Empire unless a specific tile conversion says otherwise;
- Credits are not wild workers by default;
- Yellow is Alien-specific by default.

The goal is not to make a new Roll for the Galaxy game. It is to keep tile building, goods, shipping, credits, and tableau scoring intact while removing the roll-and-reassign luck layer.

For the first playtest, do not add general free recharge beyond printed Cup placement, tile-specific retuning, or a new dummy-player system. Those are tuning knobs after the core battery loop is tested at the table.

There is also a parked mancala dice alternative in `docs/roll_mancala_dice_design.md`. It uses the base game's physical dice as mancala stones. That design is not part of this ruleset.

Components

Use all normal Roll for the Galaxy components except the Dice Cups.

Add one Phase Battery board per player. Each board has tracks from 0 to 6:

Explore	Blue	0	1	2	3	4	5	6
Develop	Brown	0	1	2	3	4	5	6
Military	Red	0	1	2	3	4	5	6
Settle	White	0	1	2	3	4	5	6
Produce	Green	0	1	2	3	4	5	6
Ship	Purple	0	1	2	3	4	5	6
Alien	Yellow	0	1	2	3	4	5	6

Blue, Brown, Red, White, Green, Purple, and Yellow have two values:

- Current pips: how many pips are available to spend now.
- Max pips: the highest that color can recharge to.

Red pips are Military value/readiness. Settling a Military World checks Red against the World's Military cost, then exhausts 1 current Red. Red current can be recharged during Manage Empire like other tracks.

White pips are Settle workers for non-Military Worlds. Credits are unlimited chips, tracked as a separate chip count instead of on a track.

Recommended physical tracking:

- For phase batteries, put the colored die on the track to show current pips.
- Put a cube or marker on the track to show max pips.
- For Red, the max marker shows Military level and the die shows current readiness.
- Keep Credit chips beside the board.

Example:

```
Red track: 0 1 2 3 4 5 6
           D  M
```

The red max marker is on 4, so Red Military level is 4. The red die is on 3, so you have 3 current Red readiness pips.

Color Map

Use these phase mappings:

Blue	Explore
Brown	Develop
Red	Military level/readiness
White	Settle non-Military Worlds
Green	Produce
Purple	Ship
Yellow	Alien

Yellow is not a normal phase. It starts at 0/0 and is used only for Alien work, explained under Yellow / Alien Pips.

Setup

Set up the base game normally, with these changes:

1. Do not give players Dice Cups.
2. Do not put dice in the Cup or Citizenry.
3. Give each player a Phase Battery board.
4. Set starting tracks:

Blue	3/3
Brown	3/3
Red	3/3 Military readiness/level
White	3/3
Green	3/3
Purple	3/3
Credits	1 unlimited chip
Yellow	0/0

Older examples below may still use 2/2 as small numbers for illustration.

Then set up starting tiles normally:

1. Give each player one Faction tile.
2. Give each player one Home World tile.
3. Apply starting die gains using their printed locations. Cup pips begin ready; Citizenry and World pips increase max only.
4. Draw starting Game Tiles into the Construction Zone as usual: one Development and one World.
5. Remove six-cost Developments from the normal tile bag. Reveal or set aside $2 + \text{player count}$ six-cost Developments as the end-game goal pool. Each player chooses 2 candidate goals from that pool; unused goals are left out unless an Explore effect later adds them.

Example:

Starting Red is 2 Military.
Your Home World grants one Military die to your Cup.
Military maps to Red.
Red 2/2 becomes 3/3.

If a starting tile places a die as a Good on that World, place a Good marker on that World, increase the granted color's max, and do not increase current.

Round Structure

Each round has four steps:

1. Select Phases
2. Reveal Phases
3. Resolve Selected Phases
4. Manage Empire

There is no Roll Dice step and no Assign Dice step.

During Select Phases, each player secretly selects eligible phases:

- Solo and 2-player games: each player selects 2 eligible phases.
- Games with 3 or more players: each player selects 1 eligible phase.

A phase is eligible only if you have at least 1 ready pip for that phase. For example, Explore requires at least 1 current Blue pip, Develop requires at least 1 usable Develop pip and a Development in your Construction Zone, Settle requires at least 1 usable Red or White pip and a World in your Construction Zone, and Ship requires at least 1 current Purple pip and at least 1 Good. Selecting a phase does not spend or exhaust the pip that made the phase eligible.

Reveal selected phases simultaneously. Resolve only phases selected by at least one player, in normal Roll order:

```
Explore -> Develop -> Settle -> Produce -> Ship
```

Unselected phases do not occur. During each selected phase, players act in turn order, starting with the start player and proceeding clockwise. A player may use a selected phase even if they did not select it, as long as they have usable current pips or resources for that phase.

After all selected phases resolve, each player resolves Manage Empire in turn order.

Phase Use

Spend one current pip for one worker use.

```
Spend 1 Blue pip    = one Explore worker
Spend 1 Brown pip   = one Develop worker
Spend 1 White pip   = one non-Military Settle worker
Spend 1 Green pip   = one Produce worker
Spend 1 Purple pip  = one Ship worker
```

Credits are not workers. Credits are chip resources for Trade income, recharge, and tile effects unless a specific rule calls for Credit spending. Red is not normal build currency: Military Worlds use Red as Military value, then exhaust 1 current Red as readiness after settling.

For Produce and Ship, spend native pips one at a time. Stop when you run out of pips or cannot use more of that phase. For Explore, Scout may spend multiple pips as search depth while still keeping only one tile; Stock spends pips one at a time.

For Develop, Brown pips are build workers and can be saved as progress on any Development in your Construction Zone. For Settle, non-Military Worlds spend White pips as Settle workers. Military Worlds use Red as Military value and exhaust current Red after settling. You may Settle any World in your Construction Zone that you can legally pay for.

Example Round

Your tracks:

```
Blue    1/2
Brown   0/2
Red     3/3
White   1/2
Green   2/2
Purple  1/2
```

Credits 1
Yellow 0/0

You select Settle. Other players select Develop and Ship.

Only selected phases resolve, in normal order:

Develop, Settle, Ship

You may:

- Develop by spending Brown or eligible Yellow into Developments.
- Settle a Military World only if your Red is at least its cost and you can exhaust 1 current Red after settling.
- Settle a normal World by spending White or eligible Yellow pips.
- Ship with Purple.

Explore and Produce do not occur this round because no one selected them.

After spending:

Brown 0/2
White 0/2
Purple 0/2

If a Development costs 2 and you have Brown 1/2, you may spend 1 Brown and mark 1 progress. Later, when you spend another Develop pip into that card, it completes.

Explore

When Explore occurs, spend Blue pips.

Explore can Scout or Stock.

Scout is a single-tile action. Spend one or more Explore pips, then look through that many candidate tiles of the needed type. Keep one tile and put it on the matching construction area. Return the other candidates to the tile supply.

For the prototype, each construction area can hold up to three Developments or three Worlds. Scout into the construction area with fewer cards. If both are tied, choose whether you are Scouting for a Development or a World.

If both construction areas are full, each Explore pip may instead Stock for 2 Galactic Credits.

Explore is the tempo-cost escape hatch. It does not complete tableau cards by itself, but it prevents a player from being hard-stopped by an empty Construction Zone or no Credits.

Because Scout lets you spend pips as search depth, Explore also supports strategic tile hunting. The prototype recognizes these broad lanes:

Builder	Development cards and end-game goal setup.
Settler	Worlds and die grants.
Producer	Production Worlds, Green, Purple, and trade/consume Developments.
Shipper	Purple, production support, and Phase V Developments.
Mining	Rare Element Worlds, Brown, and Mining League.
Novelty	Novelty Worlds, Blue, and Free Trade Association.
Genes	Genes Worlds and Green.
Alien	Alien Technology Worlds and Yellow.

Military	Red grants, Rebel/gray Worlds, and New Galactic Order.
Diverse	Missing World colors, System Diversification, and Galactic Exchange.

Example:

```
Blue is 2/2.
Your Development area has one tile.
Your World area is empty.
You spend 2 Blue to Scout for a World.
Look through 2 World candidates.
Keep 1 World and put it in your World construction area.
Blue becomes 0/2.
```

Develop

When Develop occurs, spend Brown pips into Developments in your Construction Zone.

Developments can hold progress. Brown pips, and eligible Yellow pips for Alien Developments, may be spent even if the Development cannot be completed this round. Mark that progress on the Development. You may split Develop pips among multiple Developments, and you may complete multiple Developments in the same Develop phase.

Payment may be any mix of pips:

- Brown pips;
- Yellow pips, if the Development is Alien-tagged.

If the Development's stored progress plus spent pips reaches the full cost, complete it. Credits are not wild Develop workers by default.

If a Development's progress reaches the full cost, complete that Development:

1. Move it to your tableau.
2. Apply any immediate converted effects.

Use normal Development costs. Six-cost Developments are removed from the normal bag and handled as end-game goals.

Example:

```
Brown is 1/2.
One Development costs 3.
Spend 1 Brown.
Mark 1 progress on the Development.
Brown becomes 0/2.
```

Example:

```
Brown is 2/2.
One Development costs 4.
Spend 2 Brown.
Mark 2 progress on the Development.
Brown becomes 0/2.
```

Later:

Brown is 1/2.
The Development has 2 progress and needs 2 more.
Spend 1 Brown now, then spend another Develop pip in a later Develop phase.
Complete the Development.

Settle

When Settle occurs, use Red for Military Worlds or White pips for normal Worlds.

You may spend Settle pips into any normal World in your Construction Zone. Normal World progress is persistent, just like Development progress. You may complete any Military World if you can meet its Military cost and exhaust 1 current Red after settling. You may complete multiple Worlds in the same Settle phase.

For normal Worlds, each World's payment may be any mix of pips:

- White pips;
- Yellow pips, if the World is Alien Technology;

For each Military World, your Red must be at least the World's cost. Then exhaust 1 current Red. The max marker does not spend down; the current Red pip does.

If a normal World's stored progress plus spent Settle pips reaches its full cost, complete the World:

1. Move it to your tableau.
2. Gain any dice listed on the World by pipping up those colors.
3. Gain any listed Galactic Credits.
4. Place a starting Good if the tile's die placement says the die begins on the World.

Example:

White is 2/2.
One normal World costs 2 and grants one Novelty die.
Spend 2 White to complete it.
Novelty maps to Blue.
Blue 2/2 becomes 3/3.
White becomes 0/2.

Example:

Red is 3/3.
One Military World costs 3 and grants one Military die.
Settle it by exhausting 1 current Red.
Red 3/3 becomes 2/3.
The World grants one Military die.
Red becomes 3/4.

Produce

When Produce occurs, spend Green pips.

Each pip produces one Good on an eligible non-gray World in your tableau.

Instead of placing a worker die as the Good, place a Good marker, cube, or spare die on the World.

The Good's color is the color of the pip spent:

```
Green pip = Green Good
Yellow pip = Yellow Good, only if using Yellow for Alien Produce
```

Each World can hold one Good unless a tile power says otherwise.

Example:

```
Green is 2/2.
You have two empty colored Worlds.
Spend 2 Green.
Place two Goods.
Green becomes 0/2.
```

Ship

When Ship occurs, spend Purple pips.

Each pip Ships one Good. For each shipped Good, choose Trade or Consume.

Trade removes the Good and gains Galactic Credits by World/good type:

- Blue/Novelty Good: 3 Credits.
- Brown/Rare Elements Good: 4 Credits.
- Green/Genes Good: 5 Credits.
- Yellow/Alien Technology Good: 6 Credits.
- Credits are unlimited chips.

Consume removes the Good and claims VP chips from the pool. The current prototype uses the base Consume value plus converted matching bonuses:

- 1 VP chip for consuming a Good.
- +1 VP chip if the Good color matches the World color, or the Good is wild.
- +1 VP chip if the Ship pip color matches the World color, or the Ship pip is Purple.

Purple is the normal Ship color.

Example:

```
Purple is 2/2.
You have two Goods.
Spend 1 Purple to Trade one Novelty Good for 3 Credits.
Spend 1 Purple to Consume the other Good for VP chips.
Purple becomes 0/2.
```

Gaining Dice

Whenever a tile says to gain a die, increase that die color's max. The printed die location determines readiness:

```
Cup during setup:  increase max and current immediately.
Cup during play:   increase max; free-recharge that pip during Manage Empire.
Citizenry:         increase max only; recharge normally with Credits.
World as a Good:   increase max only and place a matching Good.
```


Procedure:

1. Increase that color's max by 1.
2. Apply the printed Cup, Citizenry, or World placement.
3. The max cannot exceed 6.
4. Any gain beyond 6 is lost and creates no readiness or Good.

Color conversion:

Home	White
Novelty	Blue
Rare Elements	Brown
Genes	Green
Alien Technology	Yellow
Military	Red
Consumption	Purple

Examples:

Blue is 0/2.
You gain one Novelty die to your Cup during play.
Blue becomes 0/3.
During Manage Empire, it free-recharges to 1/3.

Green is 0/2.
You gain one Genes die to your Citizenry.
Green becomes 0/3 until recharged with a Credit.

Brown is 0/2.
You gain one Rare Elements die on its World.
Brown becomes 0/3 and that World gains a Brown Good.

Blue is 6/6.
You gain one Novelty die.
Blue stays 6/6. The excess is lost.

If a tile grants two dice, increase both max tracks and apply each printed location. If both dice are the same color, increase that max twice, still capped at 6.

Military / Red Dice

In this variant, Military maps to Red pips as Military value/readiness.

- gaining a Military die follows its printed location like any other die;
- losing a Military die reduces Red max by 1 and caps current Red to the new max;
- Red is checked as the Military value to settle Military Worlds;
- settling a Military World exhausts 1 current Red after settling;
- end-game goals that count Military dice count Red max.

For the prototype, a World is Military if it grants a Red die or has Rebel identity. Normal Worlds, including non-military gray Worlds, use White Settle pips instead.

Losing Dice

Whenever a tile says to lose or remove a die, reduce one track by 1 max.

If the track's current pips are now above its max, lower current pips to the new max.

For the first playtest, choose the lost color this way:

1. If the tile specifies a die color, reduce that color.
2. If it does not specify a color, the player chooses one color with max above 0.

Example:

```
Red is 4/4.  
You must lose one Military die.  
Red becomes 3/3.
```

```
Blue is 5/5.  
You lose one Novelty die.  
Blue becomes 4/4.
```

Yellow / Alien Pips

Yellow represents Alien Technology.

Yellow starts at 0/0. When you gain an Alien Technology die, increase Yellow max and apply its printed location.

Current recommended rule:

Yellow may be spent only when the work involves an Alien tile or Alien Good.

Yellow can pay for:

- Develop, if the Development is Alien-tagged.
- Settle, if the World is Alien Technology.
- Produce, if producing on an Alien Technology World.
- Ship, if Shipping a Good from an Alien Technology World.

Yellow cannot pay for unrelated non-Alien work.

Example:

```
Yellow is 1/1.  
One World is Deserted Alien Colony.  
During Settle.  
You may spend 1 Yellow as a Settler for that World.  
Yellow becomes 0/1.
```

Example:

```
Yellow is 1/1.  
One World is Gem World.  
During Settle.  
You may not spend Yellow, because Gem World is not Alien.
```

Simple alternative:

If the Alien rule is too fiddly, treat Yellow as extra Ship capacity:

```
Ship spending order: Purple -> Yellow
```

The simulator supports both versions. The current default is Alien mode.

Galactic Credits And Recharge

Galactic Credits no longer recruit dice from Citizenry.

The main renewable Credit source is tempo: Stock through Explore after both construction areas are full, or Trade through Ship. Immediate tile effects can also grant Credits.

Credits are not wild workers by default. Develop uses Develop pips. Settle uses White pips for non-Military Worlds and Red for Military Worlds. A specific tile conversion may still grant or spend Credits if that is the closest translation of its printed effect.

Credits are unlimited chips. They are not assigned to phases. Each round, choose how many Credits to spend on Manage Empire recharge and any specific tile effects that call for Credits.

During Manage Empire:

1. Apply queued Cup-placement free recharges to their original tracks.
2. Spend Galactic Credits to recharge colored tracks.
3. Each Credit recharges 1 current pip on one track.
4. A track cannot recharge above its max.
5. You may distribute Credits among tracks however you choose.
6. If your Credit count is at 0 after Manage Empire, set it to 1 as in the base game.

There is no general free recharge in the recommended version. A die placed in the Cup during play receives its one location-based free recharge during Manage Empire.

Example:

```
You have 3 Galactic Credits.
```

```
Brown is 0/2.
```

```
Green is 1/3.
```

```
Purple is 2/2.
```

```
Spend 2 Credits: Brown 0/2 -> 2/2.
```

```
Spend 1 Credit: Green 1/3 -> 2/3.
```

```
Purple is already full, so you cannot recharge it.
```

Construction Zone And Tableau

Each player has three separate places for tiles:

- Tableau: completed tiles. Starting Faction and Home World tiles begin here.
- Development construction area: unbuilt non-6-cost Development tiles.
- World construction area: unsettled World tiles.

The two construction areas are parallel queues, not stacks. You may build any tile in the matching construction area; it does not have to be the top, newest, oldest, leftmost, or rightmost tile.

Construction area limits:

- Development construction area limit: 3 tiles.
- World construction area limit: 3 tiles.
- Six-cost Development goals are not in either construction area and do not count against either limit.

Explore adds tiles to construction areas:

- Scout for a Development: put the chosen non-6-cost Development into your Development construction area.
- Scout for a World: put the chosen World into your World construction area.
- If the matching construction area is full, you cannot add that kind of tile unless a rule first frees space.

Develop construction:

- Develop pips are spent into Development tiles in your Development construction area.
- Development progress is persistent. Mark stored progress on each Development separately.
- You may split Develop pips among multiple Development tiles.
- When a Development's stored progress reaches its cost, complete it immediately: move it from the Development construction area to your tableau and apply converted effects.
- Multiple Developments can complete in one Develop phase.
- Credits are not wild Develop workers by default.

Settle construction:

- Normal Worlds are paid with White Settle pips, plus eligible Yellow pips for Alien Technology Worlds.
- Normal World progress is persistent. Mark stored progress on each World separately.
- You may split Settle pips among multiple normal Worlds.
- Military Worlds require current Red at least equal to the World's cost, then exhaust 1 current Red after settling.
- Military Worlds do not store progress; they settle when Red is high enough.
- You may settle or progress any World in your World construction area.
- Multiple Worlds can complete in one Settle phase.
- When a World completes, move it from the World construction area to your tableau and apply converted effects.

Six-cost Development goals:

- Six-cost Developments are removed from the normal tile bag during setup.
- They are goal cards, not construction tiles.
- Candidate and committed goal cards are kept beside the player area.
- They do not count as tableau tiles, Development construction tiles, completed Developments, or 12-square end-condition tiles.
- They score only through the goal system at game end.

The tableau end condition checks only tiles in your tableau. Starting Faction and Home World tiles count normally for tableau size and scoring, as in the base game.

Reassign Powers

Reassign powers temporarily route ready pips; they never move max capacity between tracks.

When a selected destination phase resolves:

1. Choose a current pip allowed by the printed reassign power.
2. Spend it from its original color track.
3. Use it as one worker in the printed destination phase.
4. When recharged later, it returns to its original color track.

Each reassign power may be used at most once per round unless its text says otherwise. A routed pip does not make a phase eligible for secret selection; the phase still needs its native ready pip or must be selected by another player.

Reassigning Blue to an occurring Develop phase, for example, reduces current Blue by 1 and performs one Develop worker action; Brown current and max do not change.

Use the source, quantity, destination, and Dictate restrictions printed on the Development. The simulator's retained tile data records which Developments have reassign powers but not their detailed restrictions, so it approximates each non-goal reassign Development as one any-source, any-destination route per round.

Other Tile Power Conversion

Use these conversions until each tile is individually tuned:

```
Gain/recruit a die:
Increase max; if no location is given, treat it as Citizenry.

Place a die in your Cup:
Increase max and free-recharge it during Manage Empire.

Place a die in your Citizenry:
Increase max only.

Place a die on a World as a Good:
Increase max only and place a matching Good on that World.

Return a used worker to your Cup instead of Citizenry:
Queue one free recharge of that worker's original color for Manage Empire.

Act as if you have an extra worker:
Gain 1 temporary pip in that phase this round.

Reduce a Develop or Settle cost:
Reduce the pips required by that amount.

Gain Credits:
Gain Galactic Credits normally.
```

End Of Game And Scoring

Use normal Roll for the Galaxy end conditions:

- any player has 12 or more tile squares in their tableau.
- the VP chip pool is empty.

Finish the round, then score normally:

- World and Development VP;
- VP chips;
- chosen six-cost Development goal results.

Six-cost Developments are not built from the normal bag. They are delayed end-game goals.

Setup:

1. Remove six-cost Developments from the normal bag.
2. Set aside $2 + \text{player count}$ six-cost Developments as the goal pool.
3. Each player chooses 2 six-cost Development goal candidates.

Commit trigger:

- once half of the VP chips are gone; or
- once any player has 6 completed Developments/Worlds.

When either trigger occurs, each player chooses one of their two candidate goals as their committed end-game goal. The other candidate goes back into the Development market row, or its reverse side may be used as a World.

Explore option:

- as an Explore option, a player may take another six-cost Development goal.

Scoring:

- score committed six-cost Development goals only if their minimum condition is met;
- lose 6 VP for each chosen Development goal whose minimum condition is not fulfilled.

In solo, each six-cost Development goal is tied to a named solo condition. To fulfill the goal, meet that named condition's non-score requirement. Then score the six-cost Development's converted bonus. If you miss the named condition, score -6 VP for that committed goal.

Free Trade Association	Novelty: 2+ Novelty Worlds.
Galactic Bankers	Satisfied Populace: ship 4+ Goods.
Galactic Exchange	Alien Contact: 1+ Alien World.
Galactic Federation	Developer: 4+ Developments.
Galactic Renaissance	Builder: 8+ completed tiles.
Galactic Reserves	Industrial: difficulty's Industrial max-pip target.
Mining League	Rare Elements: 2+ Rare Worlds.
New Economy	Production: 4+ production Worlds.
New Galactic Order	Military: Red max 5+.
System Diversification	Diverse: 4+ World colors.

Open playtest wording:

- clarify whether multiple players may choose the same goal from the 2 + player count pool, since that pool is smaller than two unique goals per player at 3p+.

Game Length Tuning

The tableau end condition remains the normal Roll for the Galaxy condition: 12 or more tile squares in one player's tableau. Do not raise this for the first playtest.

The current active version uses a VP chip pool. Produce creates Goods, and Ship can Trade for Credits or Consume for VP chips. Game length is governed by the 12-square tableau end condition, VP pool pressure, the recharge economy, and any solo round limit.

Solo Mode: Win Conditions

Solo mode uses the normal Phase Battery rules with a fixed clock, dummy phase cards, and win conditions. This follows the solo style of New Frontiers: Starry Rift and Jump Drive: Terminal Velocity: you are not racing an automa score. You are trying to build the right empire before time runs out.

Additional components:

- five Dummy phase cards: Explore, Develop, Settle, Produce, Ship;
- one Dummy claimed-tile row;
- one Dummy Goods marker;

- five solo campaign sheets.

Setup:

1. Set up one normal player area.
2. Set Dummy Goods to 0.
3. Shuffle the five Dummy phase cards into a face-down deck.
4. Choose one campaign sheet.
5. Use a 24 VP-chip pool: 12 for you and 12 for the Dummy seat.
6. Play exactly 15 rounds, unless your tableau reaches 12 or more tile squares first.

Each campaign sheet has four win conditions. Play four games in a row. At the end of each game, mark exactly one satisfied condition on that campaign sheet that you have not already marked.

Some named conditions are tied to a specific six-cost Development goal. To mark one of those conditions, you must:

1. meet the condition's VP threshold;
2. have committed the matching six-cost Development goal this game;
3. fulfill that goal's named-condition requirement.

The committed six-cost goal also scores normally: if fulfilled, score its converted bonus; if missed, lose 6 VP. Score-only conditions do not require a six-cost Development. Colonizer and Discovery are not currently tied to a six-cost Development and use only their printed condition.

If you cannot mark a new condition, you lose the campaign. If you mark all four conditions after four successive games, you win the campaign.

Choose a solo difficulty before the first game. These first-pass thresholds are calibrated for the 15-round solo cap, two player phase selections, two Dummy phase cards, starting colored tracks at 3/3, and a 24-chip solo VP pool:

Difficulty	Great	Triumphant	Epic	Named	Industrial
Easy	24+	30+	36+	26+	17+ max pips
Normal	30+	36+	42+	32+	19+ max pips
Advanced	36+	42+	48+	38+	21+ max pips
Very Hard	42+	48+	54+	44+	23+ max pips

Named win conditions:

Great	Score the difficulty's Great VP. No six-cost goal required.
Triumphant	Score the difficulty's Triumphant VP. No six-cost goal required.
Epic	Score the difficulty's Epic VP. No six-cost goal required.
Builder	Named VP, Galactic Renaissance, and complete 8+ tiles.
Developer	Named VP, Galactic Federation, and have 4+ Developments.
Colonizer	Named VP and have 5+ Worlds. No six-cost goal required.
Satisfied Populace	Named VP, Galactic Bankers, and ship 4+ Goods this game.
Industrial	Named VP, Galactic Reserves, and have the difficulty's Industrial max pips.
Production	Named VP, New Economy, and have 4+ production Worlds.
Diverse	Named VP, System Diversification, and have 4+ different World colors.
Novelty	Named VP, Free Trade Association, and have 2+ Novelty Worlds.
Rare Elements	Named VP, Mining League, and have 2+ Rare Worlds.
Alien Contact	Named VP, Galactic Exchange, and have 1+ Alien World.

Military	Named VP, New Galactic Order, and have Red 5.
Discovery	Named VP and have Blue max 5. No six-cost goal required.

Campaign sheets:

Outreach
Great, Colonizer, Builder / Galactic Renaissance, Discovery.

Industrial Base
Triumphant, Developer / Galactic Federation, Industrial / Galactic Reserves,
Production / New Economy.

Sector Survey
Great, Diverse / System Diversification, Novelty / Free Trade Association,
Rare Elements / Mining League.

Alien Contact
Triumphant, Alien Contact / Galactic Exchange,
Diverse / System Diversification, Discovery.

Mastery
Epic, Military / New Galactic Order, Industrial / Galactic Reserves,
Satisfied Populace / Galactic Bankers.

For one-off practice, ignore the campaign sheet and simply note every condition you satisfied.

Solo round structure:

1. Select two player phases. You may select a phase only if you have at least 1 ready pip for that phase. Selecting a phase does not spend that pip.
2. Reveal two Dummy phase cards. If the deck is empty, shuffle all five Dummy phase cards to form a new deck first.
3. The selected phase set is your phases plus the Dummy phases.
4. Resolve selected phases in normal order: Explore, Develop, Settle, Produce, Ship.
5. During each selected phase, you may spend usable pips for that phase. If the Dummy selected that phase, also resolve that Dummy effect.
6. Manage Empire normally, then check whether the six-cost goal commit trigger has been reached.

The Dummy never spends pips, never has Credits, never builds an empire, never scores, and never uses tile powers. It exists to add phase uncertainty, create piggyback opportunities, and churn the tile bag.

Dummy card effects:

Explore: Claim 1 Development tile and 1 World tile from the bag.
Develop: Claim 1 Development tile from the bag.
Settle: Claim 1 World tile from the bag.
Produce: Dummy Goods +1, max 4.
Ship: Set Dummy Goods to 0.

When the Dummy claims a tile, draw the first matching tile from the bag and place it face down in the Dummy row. Do not read or resolve the tile. If no matching tile remains in the bag, skip that claim.

The Dummy's claimed tiles do not trigger the 12-tile tableau end condition. They exist only to keep the bag moving at a multiplayer-like pace.

Solo end conditions:

- the 15-round limit is reached;
- your tableau reaches 12 or more tile squares.

Finish the round, then check your campaign sheet. Mark one satisfied unmarked condition. If you cannot mark one, the campaign is lost.

Solo example:

Campaign: Outreach.

Unmarked conditions: Great, Colonizer, Builder, Discovery.

On Normal difficulty, you finish with 35 VP, 8 completed tiles, and Blue max 5.

You may mark Great, Builder, or Discovery, but not Colonizer unless you also have 5+ Worlds. Mark exactly one condition.

You select Develop. Then reveal two Dummy cards: Produce and Develop.

Selected phases are Develop and Produce.

During Develop, you may spend Brown pips. Because the Dummy also selected Develop, draw one Development tile from the bag and place it face down in the Dummy row.

During Produce, you may spend Green pips. Because the Dummy selected Produce, Dummy Goods 0 -> 1.

Later:

Dummy reveals Explore and Ship.

Explore: draw one Development and one World from the bag into the Dummy row.

Ship: Dummy has 1 Good, so set Dummy Goods to 0.

Full Round Example

Initial tracks:

Blue	2/2
Brown	2/2
Red	2/2
White	2/2
Green	2/2
Purple	2/2
Credits	1
Yellow	0/0

Your Home World grants one Military die:

Red 2/2 -> 3/3

Round 1 phase order:

Explore, Develop, Settle, Produce, Ship

Explore:

Spend 2 Blue to Scout for a World.
Look through 2 World candidates.
Keep 1 World.
Blue 2/2 -> 0/2.

Develop:

Spend 2 Brown on a cost-2 Development.
Brown 2/2 -> 0/2.
Complete the Development.

Settle:

One World is a cost-3 Military World.
Red is 3/3.
Settle it by exhausting 1 current Red.
Red 3/3 -> 2/3.
Complete the World.
The World grants one Genes die.
Green 2/2 -> 3/3.

Manage Empire:

Credits 1.
Recharge Brown 0/2 -> 1/2.
Credits would become 0, so reset to 1.

End of round tracks:

Blue 1/2
Brown 1/2
Red 2/3
White 2/2
Green 3/3
Purple 2/2
Credits 1
Yellow 0/0

Playtest Defaults

Use these defaults:

Starting main tracks: 3/3, including White
Starting Credits: 1 unlimited chip

Starting Yellow:	0/0
Track max:	6
Credits:	unlimited chips
VP pool:	7 per player
Solo VP pool:	24 total chips
Scoring:	tableau VP + VP chips + chosen 6-cost goals
Free recharge:	0
Yellow mode:	Alien

Current playtest note:

Red readiness is a soft Military brake, not a hard cooldown. In a focused stress pass, a Military player with enough Red could still settle a Military World every round by preserving or generating enough Credit to recharge current Red during Manage Empire. That is acceptable for the first table test because it makes Military compete with other recharge and build spending instead of making Military Worlds free forever.

Watch these table metrics:

- Military Worlds completed per player.
- Rounds where Military settled while spending 0 Credits.
- Credits spent on Red recharge.
- Red at game end.
- Whether normal Worlds' Produce/Ship payoff keeps up with Military tempo.

If Military still runs away, the next clean tuning knob is to make Military Worlds that grant Red increase Red max only; their current Red gain would wait until a later recharge.